

# The cut-off mirrored double cone

*Stage 1 as an hourglass — and a cone the model shapes itself*

E-080 • design proposal • 15 September 2026 • **nothing here is measured yet**

The deck turns Chris's four points into a specification: a waist instead of a tip, a mirrored cone of targets, a random mix of four tasks, and a cone whose shape the model learns per region.



# One cone, five seeds, one clear reading

## E-069 • the cone-native codec

- 30 days of dots around a North Atlantic cell, one 3×3 patch at the present, targets at +1 and +2 pentads
- History did **not** put the cell's own velocity into the embedding — five seeds, two objectives, refuted
- History **did** buy persistent fields (hidden sea-surface height at ~0.36 of the bar vs 0.75–0.90) and a 7–8 % better one- and two-pentad forecast

## E-071 • cone v2

- Global, harmonic climatology, profile tokens — the geometry's known faults, fixed
- Reach from the fastest mechanism × 1.5 — a hemisphere by lag 3, so dots go log-radial

**The open question v2 left:** a cone wide enough to never exclude a driver samples the near field at one dot per thousands of kilometres.

# Past cone → waist → future cone



- **Past cone** (lags  $-6 \dots -1$ ): read as input, dots at the per-lag ellipse plus the anchor column
- **Waist** (lag 0): a disc of several cells — the cut-off tip — read as input and reconstructed; the embedding  $z$  summarises the waist
- **Future cone** (leads  $+1 \dots +6$ ): never read; sampled the same way and used as targets only
- **Point-mirrored**: the future cone at lead  $+\ell$  is the past cone at lag  $-\ell$  reflected through the anchor

# A gradient does not exist within a single cell

- The waist is a disc of radius  $r_0$  around the anchor, sampled as a small sunflower: ~13 cells at the default (one centre + 12), never fewer than the 3×3 patch's nine
- Read as input at lag 0 (values + observed flags) and reconstructed as the anchor family's target
- **What the waist buys:** thermal wind is a density gradient; geostrophy is a sea-surface-height slope — both need neighbours at the same instant, not in the past
- The waist has the same shape parameters as the cone at lag 0 (orientation, two semi-axes), so a boundary current can have an elongated waist along the flow
- The embedding  $z$  summarises the waist; one  $z$  per anchor, as today

≈ 13

**cells at the waist**

one centre + 12, never fewer than the nine of the 3×3 patch

1  $z$

**per anchor, as today**

so stage 2 and every archived probe see the same kind of object

# Sampled like the past cone, read by nobody

## +1 ... +6 pentads

30 days ahead, matching the 30 days of history

## targets only

nothing in the future cone is ever read by the encoder

## frozen at evaluation

the fixed v2 hourglass scores every arm and every seed

- Leads +1...+6 pentads (30 days ahead), the mirror image of the past cone through the anchor: centre  $c(+\ell) = -c(-\ell)$ , same axes and orientation at the same  $|\ell|$
- Dots are **targets only**: each is a decoder query (channel,  $\Delta x$ ,  $\Delta y$ , lead, depth)  $\rightarrow (\mu, \log \sigma^2)$ , scored with the Gaussian negative log-likelihood already in `cone_codec.py`
- The anchor column at every lead is always among the targets — the fixed-frame forecast stage 2 needs
- Per-lead and per-family loss, each reported against its own predict-the-mean bar (the E-069b rule)
- Evaluation geometry is **frozen** (the fixed v2 hourglass), whatever the training cone learns — so every number stays comparable across arms and seeds

# One model, four questions, drawn at random per example

| task                    | input                                           | targets                         | share | what it teaches                                            |
|-------------------------|-------------------------------------------------|---------------------------------|-------|------------------------------------------------------------|
| T1 forecast             | waist + past cone                               | future cone                     | 0.35  | the full problem                                           |
| T2 snapshot forecast    | waist only (past cone fully hidden)             | future cone                     | 0.15  | the built-in twin: what history buys, inside one model     |
| T3 nowcast from history | past cone under a dropout pattern; waist hidden | waist                           | 0.25  | the present from the past — persistence, advection, memory |
| T4 fill-in              | waist + past cone, both under a dropout pattern | the hidden waist cells and dots | 0.25  | E-069b's reconstruction family, kept                       |

Dropout patterns for T3 and T4, drawn independently (each fires with its own probability)

**channel drop**

at the waist — lag 0 only,  
chan\_drop\_scope lag0

**recent-lag band**

hide every dot at  $\text{lag} \leq \ell_o, \ell_o \in \{1, 2, 3\}$

**bearing wedge**

hide a 90° sector

**random dots**

each dot hidden with  $p = 0.3$

**whole waist**

T3 only

# Three rules, all learned the hard way in E-069

## 1 • Knowable targets only

A target channel must be visible somewhere in the input (any lag, any dot). Under the first E-069 plan 80 % of hidden-dot targets broke this and 64 % of the loss was predict-the-mean.

## 2 • No copy targets

The anchor family scores hidden channels only (anchor\_hidden\_only); a target visible in the input is a copy, not a question.

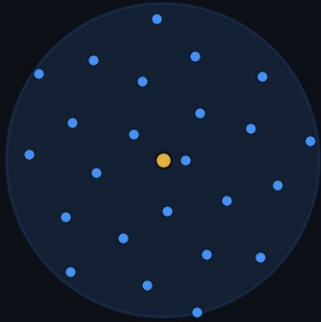
## 3 • Every family against its own bar

Per (task, family, lead) loss, each divided by its predict-the-mean score; a task is “learned” only where the ratio is below one at held-out anchors.

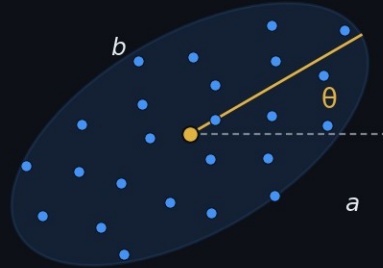
- The task is not told to the model by a token; the masks are visible (the mask\_tok / miss\_tok distinction stays), and the lead is in every query's coordinates — that is enough
- Loss weights: equal across tasks per example; the shares in the table set the mix, not the weights

# Seven numbers per region shape the whole hourglass

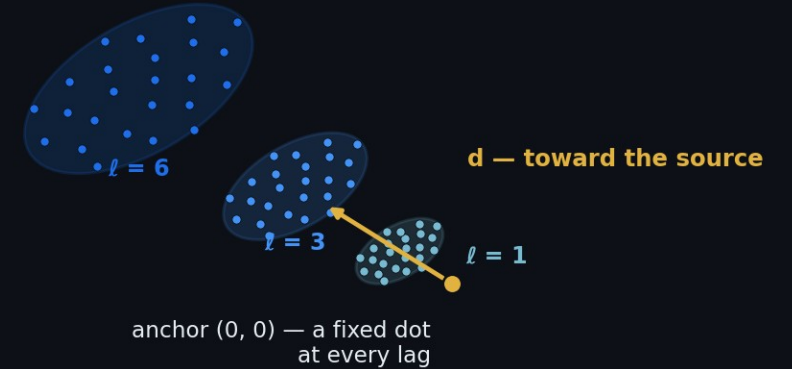
(a) canonical sunflower, unit disc  
24 points, golden angle  $137.508^\circ$



(b) rotated by  $\theta = 30^\circ$ , stretched to 2:1  
semi-axes  $a, b$



(c) the same pattern at three lags: shifted by  $\ell \cdot d$ , grown with  $\ell$



For each region and channel group, at lag  $\ell$  (past,  $\ell = 1 \dots 6$ ) the sample points are

$$p(\ell, i) = \text{anchor} + \ell \cdot d + R(\theta) \cdot \text{diag}(a_0 + \ell \cdot s_a, b_0 + \ell \cdot s_b) \cdot u_i$$

- $u_i$ : the canonical sunflower on the unit disc (Vogel spiral, golden angle), fixed, ~24 points — the same pattern at every lag
- $d$  (2 numbers): the drift per pentad, in km — where the cone's centre moves as you go back in time; this is the arrow toward the source
- $\theta$  (1): orientation of the ellipse
- $a_0, b_0$  (2): the waist semi-axes, in km — the cut-off tip's size and elongation
- $s_a, s_b$  (2): how fast each axis grows per pentad — spread
- **Future cone**:  $p(-\ell)$  reflected,  $c(+\ell) = -\ell \cdot d$ , same axes at the same  $|\ell|$ ; the anchor column  $(0, 0)$  is added at every lag regardless

# Continuous positions, interpolated reads — the gradient reaches the geometry

## Sample between the cells

A dot at a continuous position reads the field by **bilinear interpolation** over its four surrounding cells, weighting by the observed flags:  $\text{value} = \sum w_i m_i v_i / \sum w_i m_i$ ,  $\text{observed-fraction} = \sum w_i m_i$

## The gradient exists

The interpolation weights are functions of the position, the position of the seven numbers, so  $\partial \text{loss} / \partial (\mathbf{d}, \theta, a, b, s)$  exists — the deformable-convolution / spatial-transformer trick, applied to a gather instead of a convolution

## Footprint, not point

Each dot averages over a Gaussian footprint whose width tracks the local dot spacing ( $\sim r/\sqrt{n}$ ), read from a mip-map pyramid of the tensor — so a dot means “this area”, and the gradient is not confined to four cells

## Same cost class as today

One gather per dot (4 cells, or one pyramid cell), on the fly, keyed exactly as E-069's examples are — plus the pyramid, built once per tensor

# A coarse map per channel group, interpolated to the anchor

- A 2° grid over the globe (90 × 180 cells); the anchor's parameters are the bilinear blend of the four surrounding map cells — smooth by construction, no seams
- **One map per channel group**, because the source of a signal depends on what moves it
- **Optional, phase 2:** a first-harmonic seasonal term on the drift,  $d(\tau) = d_0 + d_1 \cos \tau + d_2 \sin \tau$  — the Somali Current reverses with the monsoon
- **Why per depth:** at the RAPID line the upper limb flows north and the Deep Western Boundary Current flows south — the same anchor should point its surface cone south and its deep cone north

## the six channel groups

**ocean surface** — currents, sea-surface height, sea-surface temperature, mixed layer

**Argo upper** —  $\leq 300$  dbar

**Argo mid** — 300–1000 dbar

**Argo deep** —  $> 1000$  dbar

**atmosphere** — drift only at lags 0–1

**land** — no drift:  $v = 0$ , axes and angle only

# $\approx 0.7 M$

## geometry parameters

16,200 cells × 6 groups × 7 numbers — a tenth of the 7 M codec

# Bounded by physics, tied to the fixed cone, started isotropic

## Speed cap

$|d| \leq v_{\text{design}} \cdot \Delta t = 5.4 \text{ m/s} \times 5 \text{ d} = 2,333 \text{ km}$  per pentad (E-071's fastest mechanism  $\times 1.5$ ), via a tanh parameterisation — the learned cone lives inside the v2 envelope

## Floors

$a_0, b_0 \geq \text{one cell (28 km)}$ ;  $s_a, s_b \geq 0$ ;  $a_0 + 6 s_a \leq v_{\text{design}} \cdot 6 \Delta t$

## Mirror tie, by default

The future cone shares the seven numbers (reflected); an untied arm is the ablation, not the design

## Stop-gradient on the target side

The loss reaches the geometry only through the **input** reads; the positions of targets are treated as constants

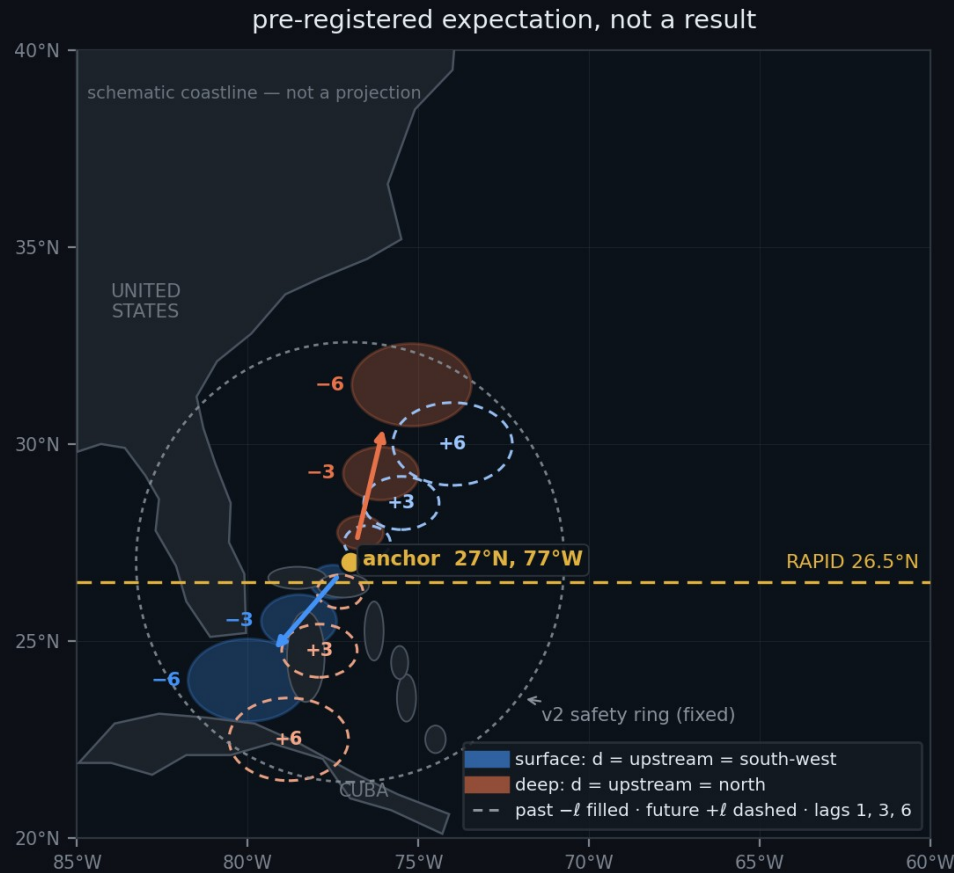
## The v2 safety net stays

In addition to the learned ellipse, each lag keeps one fixed log-radial ring of 24 dots to the antipode — the cone can be directed without asserting that a driver could not have come from elsewhere

## Init

$d = 0, \theta = 0, a_0 = b_0 = 3 \text{ cells}$  (aspect 0.71 applied),  $s$  from E-069's 0.3 m/s — the fixed hourglass exactly, so step 0 of the learned arm is the control

# One anchor on the western boundary, two cones pointing opposite ways



- **Surface group:** d points south-west (up the Florida Current, toward the Loop Current); the future cone leans north-east along the Gulf Stream
- **Deep group:** d points north along the slope (up the Deep Western Boundary Current); the future cone leans south
- Neither is programmed: these are the **pre-registered expectations** for the learned maps — the check that the mechanism works
- If the learned surface drift correlates with  $-u_{\text{clim}}$  (the climatological surface current, reversed) over the world ocean, the cone learned “upstream” from the data alone

# A learnable cone can learn to ask easy questions

**The loophole:** if target positions could move, the gradient would move them to smooth water — a lower loss that means nothing.

## Fix 1 • stop-gradient on target positions

Slide 11. The geometry is trained only by “which inputs help”, never by “which targets are easy”.

## Fix 2 • the mirror tie

The future cone cannot be tuned separately from the past cone.

## Fix 3 • frozen evaluation geometry

Held-out numbers are always scored on the fixed hourglass, for every arm.

**Residual second-order effect:** as the input cone moves, the mirrored targets move with it. Monitored, not assumed away — report the target-variance under the learned cone vs the fixed cone at every eval; a drop > 10 % is a flag.

# Tokens, gathers, parameters — the same regime as v2

|                                  | E-069 (measured)              | this design (budget)                                                                                         |
|----------------------------------|-------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>waist</b>                     | 42 patch tokens (14 ch × 3×3) | 13 per-location tokens carrying all surface channels                                                         |
| <b>past cone</b>                 | 706 dot tokens                | 6 lags × (1 + 24 ellipse + 24 ring) × 4 groups ≈ 1,176                                                       |
| <b>Argo</b>                      | 192 (32 × 6, column only)     | 26 profile tokens (E-071 §3)                                                                                 |
| <b>targets</b>                   | 42 + 84                       | 256 dot queries per example, drawn from the future cone and the hidden waist / dots (as n_dot_queries today) |
| <b>geometry parameters</b>       | 0                             | ≈ 0.7 M (slide 10)                                                                                           |
| <b>extra per-tensor artefact</b> | —                             | the mip-map pyramid (~1/3 of the tensor, built once)                                                         |

- Perceiver cost is linear in tokens: ~1.6× E-069's encoder time, in line with v2's own 2.1× estimate; the interpolated gather is 4 cells instead of 1
- Training cost per seed: E-069 seeds ran under one 4090-hour on the North Atlantic; budget 2–3 4090-hours per seed on the global tensor

# Three arms, three seeds, four read-outs, one verdict rule

## A0 • fixed

The hourglass with the v2 geometry (isotropic ellipses, ring, mirror), no learning — the control, and the eval geometry for everyone

## A1 • learned

The seven numbers per region, initialised at A0

## A2 • learned, primed

As A1, but **d** initialised at  $-u_{\text{clim}} \cdot \Delta t$  (the climatological surface current, reversed) for the ocean surface group — tests whether the gradient can find upstream on its own (A1) or only keep it (A2)

**Held fixed:** same tensor (family 7.2, global), same 7 M codec, 20 k steps, frozen protocol (train  $\leq 2020$ ; 2008–09, 2016–17, 2021–24 held out), 3 seeds each.

**R1** T1 future-cone loss per lead vs bar • **R2** the T1 – T2 gap (what history buys) • **R3** drift-vs-upstream correlation, world ocean,  $|u_{\text{clim}}| > 0.2$  m/s  
**R4** the western-boundary depth check (surface south-west, deep north) • **R5** target-variance flag (slide 13)

**Verdict:** A1 adopted if it beats A0 on R1 at every lead  $\leq 3$ , paired at all three seeds, with R5 clean. A2 > A1 with A1  $\approx$  A0 means the gradient cannot find upstream — then geometry is set from climatology, not learned. Neither  $\rightarrow$  keep A0 and move to a dynamic (per-example) cone.

# Five steps, each with its test

in ml/cone.py · ml/cone\_sampler.py · ml/cone\_codec.py · ml/train\_cone.py

- 1 Waist + future cone in the geometry (cone.py)** waist\_dots, future\_dots = reflected inner\_dots; coverage\_report extended with the mirror identity (future set = -past set) — pure numpy, CPU test
- 2 The task mix (cone\_codec.py::default\_plan → plan\_v3)** T1-T4 with shares and the five dropout patterns; test: every target's channel is visible somewhere in the input, for 10 k drawn examples, zero violations
- 3 Continuous sampling (cone\_sampler.py)** pyramid build + interpolated gather with observed-weighting; test: at integer positions with a full observed mask the read equals today's gather bit-for-bit
- 4 The parameter maps (cone\_codec.py::ConeGeometry)** 2° maps per group, tanh/softplus bounds, mirror, stop-gradient, init = A0; test: at init the dots equal step 1's fixed dots to < 0.1 cell; gradient-check on the seven numbers with finite differences
- 5 Read-outs in the trainer (train\_cone.py)** per-task bars, R3/R4 maps exported as data/cone\_drift\_<group>.json for the app's Cones tab, R5 flag; then the three arms as recipes f7l2-hourglass-{fixed,learned,primed}

*Each step lands on main with its test before the next starts; step 3 is the only one that touches the data path and is gated on the bit-identity test.*

## Six choices the deck took a position on — all reversible

### Future horizon

**+6 pentads** to mirror the past; or shorter (+1...+2 as E-069) with denser targets?

### Task shares

**0.35 / 0.15 / 0.25 / 0.25** — any of them a lever you want set differently?

### Parameter maps

**2°** grids; 1° doubles resolution at 4× the (small) parameter cost

### Argo depth groups

Split at **300 and 1000 dbar** for the Argo cones

### The v2 safety ring

**Kept** at every lag (24 tokens per lag per group); dropping it halves the past-cone tokens but restores v2's “too narrow” risk

### Seasonal drift term

**Deferred** to phase 2; or in from the start (monsoon regions)?

# Sources

## the record this deck stands on

- **E-069 • Two stencils, one cone** — ml/plans/E069\_cone\_codec.md; results in ml/EXPERIMENTS.md#e-069 (five seeds, H1 refuted; what history buys)
- **E-069b masking-plan fix** — ml/cone\_codec.py::default\_plan (chan\_drop\_scope, anchor\_hidden\_only), ml/train\_cone.py
- **E-071 • Cone v2** — ml/plans/E071\_cone\_v2.md (§3 profile tokens, §4 design speeds and log-radial dots, §4.5 the split in time, §6.4 per-location tokens)
- **E-076 • Family 8 nearest observations** — ml/plans/E076\_family8\_nearest\_observations.md (the (value,  $\Delta x$ ,  $\Delta y$ ,  $\Delta t$ ,  $n_R$ ) token)
- **The geometry as built** — ml/cone.py (FAMILIES, reach\_km, slots, inner\_dots, outer\_spiral, coverage\_report)

## and on

- **Deformable sampling:** Dai et al. 2017, Deformable Convolutional Networks; Jaderberg et al. 2015, Spatial Transformer Networks; Zhu et al. 2021, Deformable DETR
- **Currents:** E-071 §4.1's table (Somali 3.6 m/s, Gulf Stream ~2.5, DWBC 0.02–0.1 m/s); RAPID 26.5°N array for the limb directions